

Can Machine Learning Predict Drug Sensitivity in Cancer?

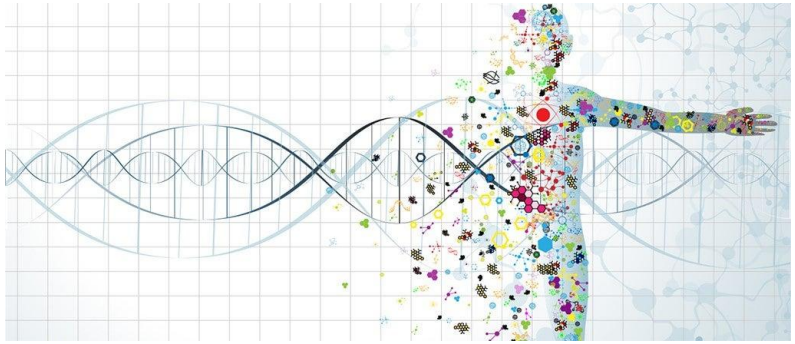
Jack Stafford

Background- Precision Medicine

Precision medicine- **The process of tailoring healthcare to a person's genetics, lifestyle, and environment.**

Could be an invaluable tool for treating many conditions- it can…

1. Improve efficacy of treatment.
2. Prevent unnecessary side effects.
3. Reduce time/money wasted on ineffective treatment.



Precision Medicine for Cancer

Cancers are a leading cause of death- responsible for roughly **10 million deaths** in 2020.

Tumours and treatment response vary greatly between patients.

It would be a **vital** area to target with precision medicine.

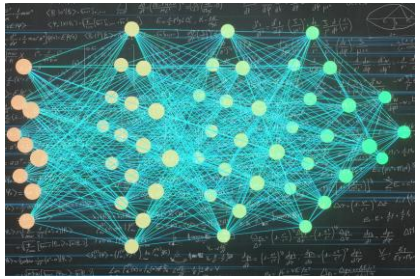
One way to do this is via predicting the response to a chemotherapy drug, but how?



Machine Learning for Response Prediction

Machine learning models are great for making predictions from available data.

They could enable prediction of drug response from biological data.



My Research Question

“Can machine learning predict cancer cell line sensitivity to a drug?”

The problem was formulated as a binary classification task.



The Dataset

I used the Genomics of Drug Sensitivity in Cancer (GDSC) dataset from Kaggle.

It contains several drug response metrics as well as biological and experimental features.

From this I chose to investigate the drug ulixertinib as:

1. It had the highest number of observations.
2. It had relatively low missing data.

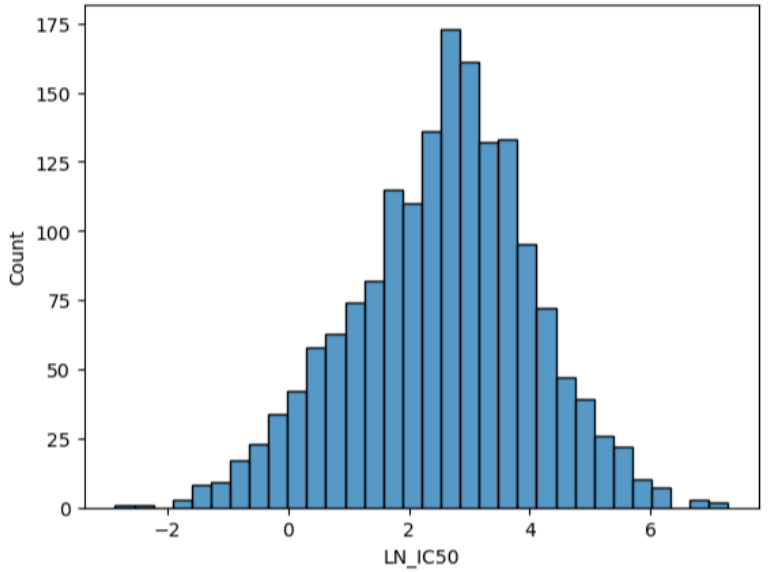
kaggle

Exploratory Data Analysis

The distribution of $\ln(IC_{50})$ (drug response metric) was visualised.

It was found to be approximately symmetric.

Missing values were assessed across features.



Target Variable

$\ln(IC_{50})$ was used as the measure of drug response.

A median split was used to create binary classes:

1. Sensitive.
2. Resistant.

This ensured a balanced class distribution.



Data Preprocessing

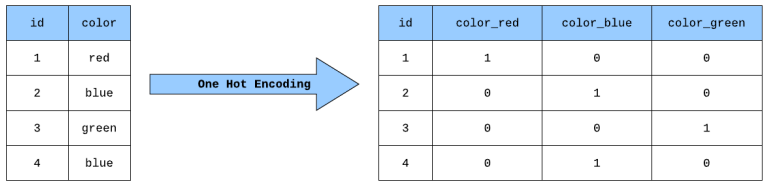
Removed…

1. Identifiers and irrelevant metadata.
2. Other response metrics (AUC and Z-score) to prevent data leakage.

Handled missing data:

1. Dropped a column with roughly 21% missing values.
2. Removed remaining incomplete rows.

One-hot encoded categorical variables.



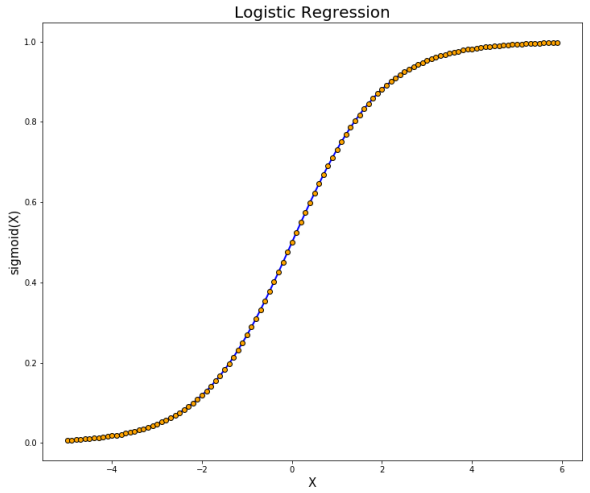
Model Training

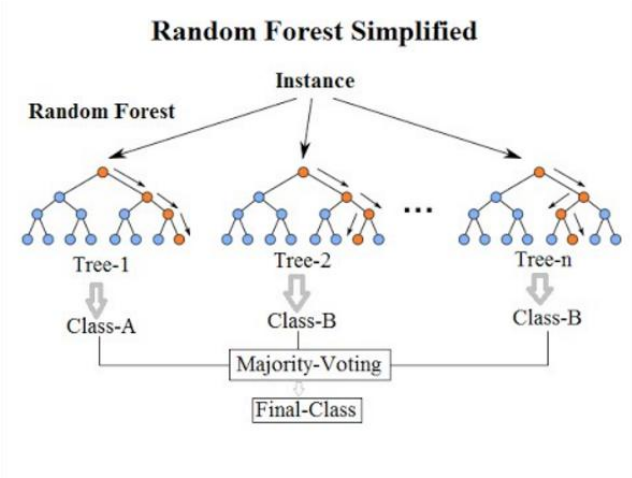
I used a train/test split of 80:20.

A logistic regression model was trained as a baseline.

A random forest model was also trained.

Default parameters were used.





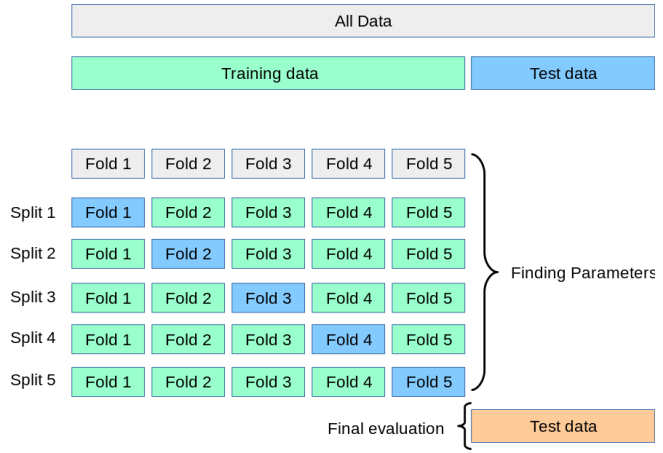
Evaluation Approach

Accuracy used as primary metric- balanced classes.

10-fold cross-validation was also performed.

Confusion matrices were used for a deeper insight.

A paired t-test was implemented to compare model performances.



Results- Accuracy

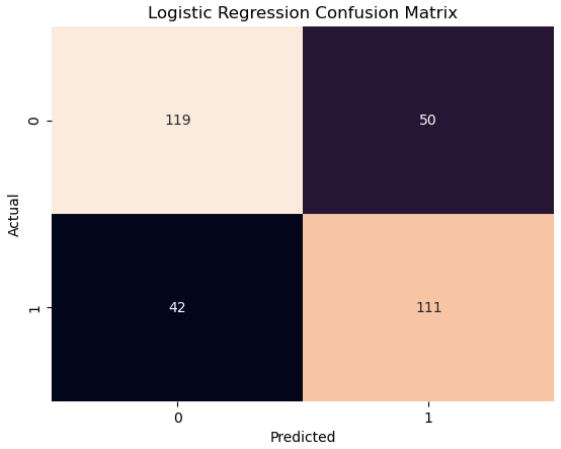
Test set accuracies:

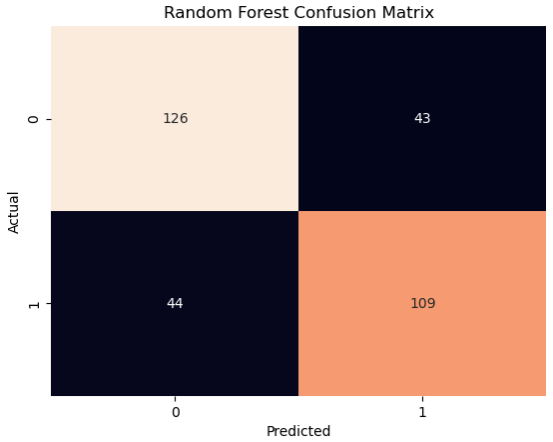
1. Logistic regression ≈ 71%
2. Random forest ≈ 73%

Cross-validation accuracies:

1. Logistic regression ≈ 66%
2. Random forest ≈ 67%

Fairly consistent across evaluation methods.





Feature Contributions

Analysed…

1. Standardised coefficients (logistic regression).
2. Feature importances (random forest).

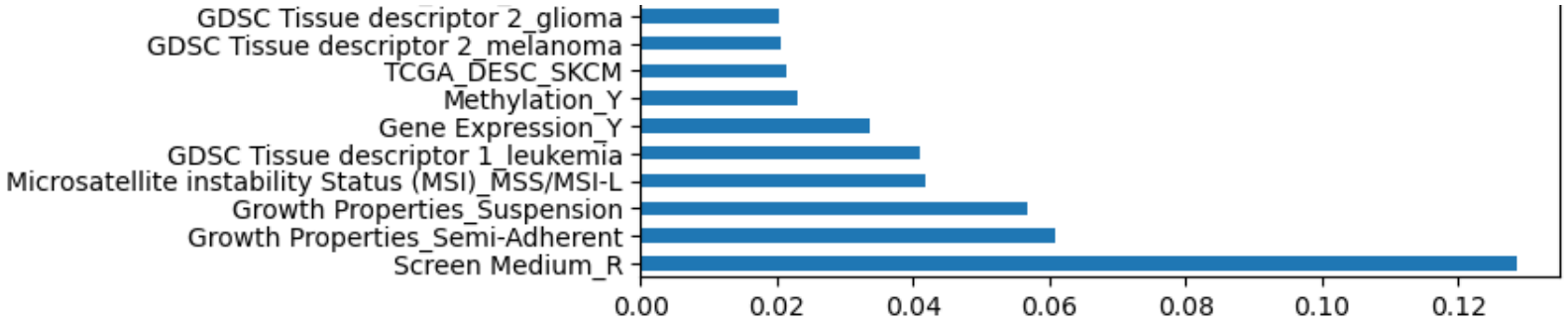
Both were a mix of biological and experimental features.

Some overlapped but there were also many distinctions.

Logistic Regression

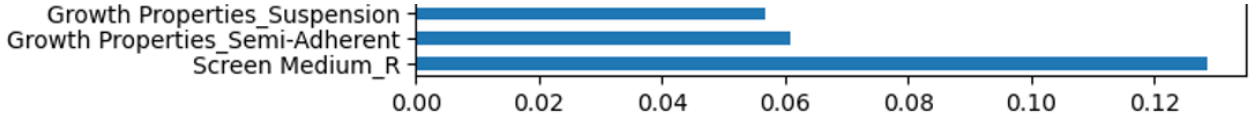
TCGA_DESC_KIRC	0.562169
GDSC Tissue descriptor 2_myeloma	0.516188
TCGA_DESC_BRCA	0.514640
TCGA_DESC_HNSC	0.494194
GDSC Tissue descriptor 2_head and neck	0.489790
Growth Properties_Semi-Adherent	0.406469
TCGA_DESC_CESC	0.358554
TCGA_DESC_PRAD	0.334834
TCGA_DESC_LUAD	0.326197
GDSC Tissue descriptor 2_leukemia	0.318908

Random Forest



Limitations

1. Binary classification oversimplifies drug response.
2. Based on in vitro cell line data (not patients).
3. Experimental features may reduce biological interpretability.



4. Moderate accuracy limits clinical application.



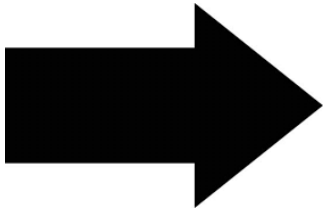
Other Discussion Points

1. Both models have similar accuracy ($p > 0.05$)- increasing complexity doesn't seem to help much.
2. Similar accuracy between test set and cross-validation accuracies- the models seem to generalise fairly well to unseen data.
3. Precision or recall could be prioritised in a clinical setting based on the treatment objective.



Areas for Future Work

1. Use real patient (in vivo) data.
2. Explore other models/techniques e.g. deep learning.
3. Try imputation methods and hyperparameter tuning.
4. Consider regression-based modelling for comparison between multiple drugs for one patient.



Conclusion

Machine learning can predict cancer cell line drug sensitivity with moderate accuracy.

Biological features of cancer cells can function as useful predictive signals.

Future work is necessary prior to clinical application.



Thank You for Listening

If you have any questions, feel free to ask.

